2023 International Conference on Mobile Internet, Cloud Computing and Information Security (MICCIS) | 979-8-3503-0093-2/23/$31.00 ©2023 IEEE | DOI: 10.1109/MICCIS58901.2023.00015

Intelligent Connected Vehicle CAN-FD Bus Network Security Protocol

1st Jingyi Jia
Yanbian University of Engineering
Yanji, China
952583702@qq.com

2nd Yihu Xu*
Yanbian University of Engineering
Yanji, China
yhxu@ybu.edu.cn

3rd Yujing Wu
Yanbian University of Engineering
Yanji, China
yjwu@ybu.edu.cn

4th Yinan Xu
Yanbian University of Engineering
Yanji, China
ynxu@ybu.edu.cn

5th Yiming Fan
Yanbian University of Engineering
Yanji, China
fanym2001@163.com

6th Dandan Zhang
Yanbian University of Engineering
Yanji, China
zddzrflcy@163.com

***Abstract*—In recent years, due to the development of intelligent connected devices, vehicle CAN-FD bus security is facing huge problems and challenges. Firstly, this paper introduces the data vertical structure, characteristics and advantages of CAN-FD bus in detail. Secondly, this paper analyzes the network vulnerability of CAN-FD bus and proposes a possible attack model against the vulnerability. Finally, on the basis of in-depth understanding of the existing defense models, this paper proposes the detection method of vehicle CAN-FD bus data and encryption authentication technology. It provides the possibility to ensure the information security of CAN-FD bus without affecting the normal running of vehicles.**

On-board CAN-FD bus; data detection; encryption and authentication; and information security.

I. INTRODUCTION

The multi-function connection between automobile and smart device can provide more convenience and a better driving experience for users, but it also brings a lot of intrusion access for malicious attackers [1]. The CAN-FD bus was considered a closed and isolated network at the beginning of its design, and all ECUs were considered legitimate and credible. In the process of frequent interaction between the vehicle network and the outside world, it increases the possibility of intrusion. Once hackers break into these systems and can successfully connect to the CAN-FD bus network level in the car, it means that the driver may have lost control of the car [2,3]. Therefore, the security of CAN-FD bus is particularly important in the running of vehicles.

II. OVERVIEW OF THE CAN-FD BUS

A. CAN bus network brief introduction

Bosch released CAN-FD (CAN With Flexible Data-Rate) scheme, which inherits the main features of traditional CAN bus, such as the use of less modified physical layer, two-wire serial communication protocol, based on non-destructive arbitration technology, distributed real-time control, reliable error handling and detection mechanism. At the same time, CAN-FD makes up for the lack of bandwidth and data length.

B. Communication frame structure of the CAN-FD bus

The standard and extended framework of CAN-FD is shown below, The CAN-FD data frame from the Frame Start, Arbitration Field, Control Field, Data Field, CRC (Stuff Count + CRC Sequence), Acknowledges Field and EOF, End Of Frame Seven different bit fields of the composition.

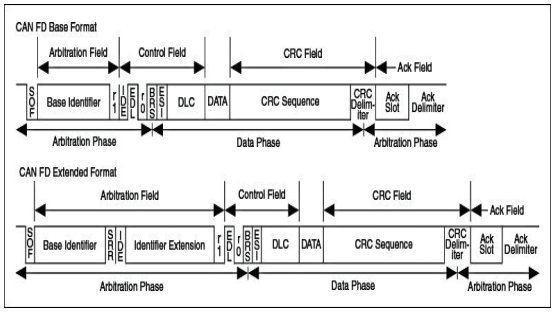


Figure 1. CAN-FD Basic and Extended Format

1) Frame start: It represents the start of data frames and remote frames, and consists of a single dominant bit. For the standard frame and extended frame, the first part sent from the node is the frame start, the SOF shows the dominant state, and all nodes must be synchronized to the frame start in front of the node that started sending the message first.

2) Arbitration field: The structure of the standard frame is different from the extended frame. In a standard format, the arbitration field consists of an identifier and an RTR bit (CAN format) or an r1 bit (CAN-FD format) with an identifier of 11 bits long. In an extended format, the arbitration field consists of an extended identifier,

979-8-3503-0093-2/23/$31.00 ©2023 IEEE
DOI 10.1109/MICCIS58901.2023.00015

Authorized licensed use limited to: Notre Dame University - Louaize. Downloaded on September 21,2026 at 19:39:41 UTC from IEEE Xplore. Restrictions apply.

SRR bit, IDE bit, and RTR bit (CAN format) or r1 bit (CAN-FD format).

3) Control field: It consists of IDE bit, FDF bit, RES bit, BRS bit (Bit Rate Switch, bit rate switch) and ESI bit (Error State Indicator, error status indication). Compared with conventional CAN, CAN-FD newly adds FDF bit, BRS bit, and ESI bit to the control field. Both CAN and CAN-FD have a 4-bit configuration, with the IDE bit representing the extended frame flag and the res bit as reserved bits.

4) Data field: The data field consists of the data to be transmitted within the data frame. It can contain 0 to 8 bytes in CAN format and 0 to 64 bytes in CAN-FD format. Each byte contains 8 bits, first sending the highest effective bit (MSB). When the DLC is zero, or in a remote frame, there is no DATA field.

5) CRC check field: It contains CRC sequence followed by recessive CRC separator. Specifically, the frame check sequence in the CRC sequence comes from a circular redundant code (BCH code), and the separator can be composed of one or two recessive bits.

6) ACK response field: It includes response gap (ACK Slot) and response definition (ACK Delimiter). When switching from a high-speed data field to a slow arbitration field, it causes the transceiver phase shift and the bus propagation delay. To compensate for its phase shift and delay, this extra 1-bit time was added to the CAN-FD. After the ACK, ACK Delimiter is sent. This is a separator indicating the end of the ACK, which is a 1-bit recessive bit.

7) Frame ending: Each data frame is defined by a landmark sequence, which consists of seven "recessive" bits.

III. Security Issues Facing the CAN-FD

A. Vulnerability analysis

Just like CAN, CAN-FD cannot provide security functions for secure communication. According to the way of CAN-FD transmission, its vulnerabilities can be divided into the following three aspects: weak access control, lack of encryption and lack of message authentication.

1) Weak access control means that in the process of physical communication, attackers can steal data information, CAN-FD bus and other bus message information through the external OBD interface (on-board diagnosis), and send illegal message information to change and control the ECU in the vehicle, threatening the safety of drivers and passengers.

2) The lack of encryption means that based on the nature of CAN-FD, all data frames are broadcast and attackers can easily eavesdrop on CAN-FD data frames. At the same time, it is very easy to analyze the meaning of the relevant data framework, and attackers can use automotive diagnostic tools to obtain the data framework that forces ECU control.

3) The lack of message authentication means that the receiver recognizes the control data frame in CAN-FD only based on the sender ID in the data frame, and the attacker can resend a valid control data frame disguised as a legitimate sender. Current CAN-FD provides only one CRC code to ensure transmission error detection.

B. Attack model

1) Stealing data: Since the CAN-FD bus uses the broadcast communication in the data link layer to transmit messages and has no authentication mechanism, hackers can easily use this defect to obtain user's privacy data such as vehicle driving track data, user identity information, vehicle mileage number and other information.

2) Injecting malicious data: Because the data sent on the CAN-FD bus lacks the data authentication mechanism, the attacker can obtain the administrator permission of the vehicle terminal by analyzing the stolen information, and send malicious messages on the bus by using the invaded or unauthorized nodes to cause confusion. An attacker can easily create and inject diagnostic messages on the bus that can cause the ECU to perform arbitrary actions, thereby taking control of various ECU components in the vehicle and influencing vehicle behavior.

3) Attacking with flood data: The attacker sends messages at a high rate within a certain period to block normal transmission frames, effectively preventing it from receiving or processing information from legitimate users, so that the bus load will reach an outlier. Flooding attacks have different effects when packets of different periods are sent. If the frequency of sending packets is higher than that of normal transmission, nodes with lower priorities will wait until the bus is idle before sending the packets. As a result, the bus transmission breaks down. When the sending frequency is low, the receiving node will perform operations according to the malicious message, and the vehicle status is completely controlled by the attacker.

4) Replaying the data: Due to the lack of freshness protection of CAN protocol, an attacker can read messages sent on the bus, and then replay them on the bus to overwrite normal data, and even control the parts of the vehicle, such as the lights or doors are always on or off, etc. If an attacker masquerades as a real node in the bus, the attacker sends packets with high-priority ids that should not appear in the current state. These packets obtain the right to send messages first, causing a large number of communication resources to be wasted.

C. Vehicle network security way

1) PKI Authentication System: The main function of this system is to bind the identity of the certificate holder and related key pairs (through issuing digital certificates for public keys and related user identity information), to provide users with a convenient way to apply for certificates, invalidate certificates, obtain certificate status inquiries, and to realize the integrity, non-repudiation and confidentiality of identity authentication of various entities in communication by using digital certificates and various related services (certificate issuing blacklist issuing timestamp service, etc.) [4].

Authorized licensed use limited to: Notre Dame University - Louaize. Downloaded on September 21,2026 at 19:39:41 UTC from IEEE Xplore. Restrictions apply.

2) Research Based on Data Encryption and Gateway Authentication: Groza et al. proposed a series of lightweight broadcast authentication protocols and carried out research on the security of vehicle CAN bus in many aspects. For example, EPSB was used to verify the availability of these protocols in the scenario where the number of ECU nodes is relatively small. Wolf et al. put forward a method of equipment authentication by issuing certificate by gateway, which issues a certificate of gateway to each control unit to carry out authentication. The control units that complete authentication with each other will accept the symmetric encryption key of the other party to carry out encrypted data transmission when transmitting data [5,6].

IV. PROPOSED SECURITY METHOD

Due to the limitations of the computing speed of the vehicle bus network, these methods can only apply extremely simple encryption methods. From today's computer security history, simple internal encryption methods are easy to crack under the malicious operation of intentional people, which is also the reason why computer viruses were once popular. Eventually, simple encryption algorithms will certainly be replaced by powerful firewall systems.

According to the common CAN-FD bus attack model, the security architecture of information anomaly detection and encryption authentication is designed. It mainly includes three parts: intrusion detection,data encryption and authentication technology. The overall module diagram is shown in figure.

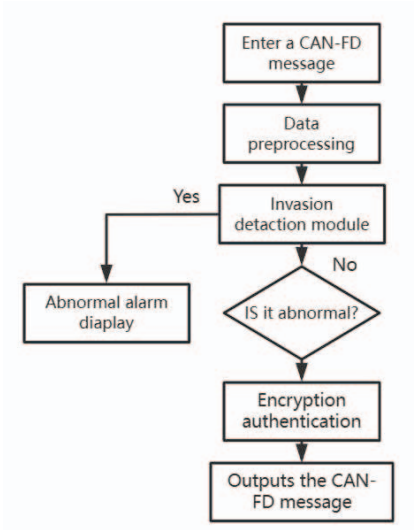


Figure 2. Overall Module Diagram

A. Intrusion detection

1) Three General Methods of Data Detection

a) Based on statistical data： By collecting a large number of historical packet records, CAN data traffic is captured, and a summary model can be established to express its random behavior. The original method was based on the univariate probability model, which was abstracted as an independent Gaussian variable and then developed a multivariable model, considering the correlation between multiple variables and the time series model to analyze the law of data change over time. The advantage of the statistics-based method is that the system does not need to know the prior elements of the attack, and can update and detect the latest attack behavior in real-time, and obtain the user's normal behavior data according to the previous data. The disadvantages are that the high dimensionality of data detection performance of the statistics-based method is poor, and the setting of exception threshold affects the detection performance.

b) Rule-based approach： Under the condition of known prior knowledge, the mode of normal data and abnormal data is artificially divided, and the data meeting the normal mode is normal data. The rule-based method can achieve a better classification effect with high robustness, but the process of detection and decision making is extremely complex, and the detection conclusion often depends on the expert's business decision level and ability, and requires a lot of manpower.

c) Method based on machine learning: Machine learning is a commonly used anomaly detection technology. Machine learning is a technology that enables computers to simulate human learning behavior, acquire new knowledge and skills through learning, and reconstruct existing knowledge structures and improve performance.

2) Data detection for the different intrusion invasions

The CAN-FD bus data field is fixed and has a maximum payload of 64 bytes. The analysis of bus message data shows that the change difference of the data sent by the context in the bus is small and relevant, so the intrusion detection algorithm we designed is more concise.

When data is transmitted on the bus, the mode of TX and RX authentication codes on the sender and receiver in the proposed architecture is the same and synchronous. It is proposed that the receiver verifies the availability of the message according to the message authentication code. The receiver uses array storage to store the packets with authentication codes sent from the sender. According to the message preprocessing module mentioned above, we designed a data array of length 50. At the same time, the counter principle is used. After verification, the counter will be increased by 1. If it passes the authentication, it receives the message and sends data as the original message to ensure the normal communication of the bus. Otherwise, the system determines the attack type based on the attack type and alerts the system to the intrusion.

For different attack types, the specific detection methods are as follows:

a) For injecting malicious data: The transmission mode of CAN-FD bus is broadcast. Hackers CAN send malicious messages to the bus and ECU with corresponding ids can receive the messages sent by hackers. Each ID is assigned a message authentication code each time a message is sent. The validity of the message can be detected by verifying the corresponding message authentication code. When the RX authentication

Authorized licensed use limited to: Notre Dame University - Louaize. Downloaded on September 21,2026 at 19:39:41 UTC from IEEE Xplore. Restrictions apply.

of the receiver is inconsistent, it is determined as an injection attack and reminds us of an injection attack.

b) For replaying data: Part of the packets in the past period are extracted for replay operation. When the RX authentication code of the receiver is the same as that of the previous part, it is judged to be subjected to a Replay attack and reminds us of the replay attack at this time.

c) For snooping data: After TX sends a message with the loaded authentication code to the bus, RX checks whether the authentication code of the message is the same after receiving the message from the sender. If part of the RX authentication code disappears, the receiving end determines that the message is under snooping attack and notifies that it is under a replay attack.

d) For flood data: In the event of a flood attack, you can check the availability of the message according to the message authentication code. If the RX authentication code on the receiving end is repeated N times and the bus load increases rapidly, the system detects a flood attack and alerts the recipient of the flood attack.

B. Encryption technology

The essence of encryption algorithm is to change the data or files that people can understand into an irregular "garbled code" according to a certain algorithm through a specific operation, which is called ciphertext The ciphertext can only display the original content after entering the correct key and then decrypting it accordingly [7]. In this way, we can protect data from being read and stolen by unauthorized people Even if the data leaks to the outside world, the outside world cannot interpret it Macroscopically, encryption algorithms can be divided into symmetric encryption technology and asymmetric encryption technology.

In vehicular networks, the communication between ECUs on vehicles is carried out in both sender and receiver, and CAN-FD bus network has high real-time requirements, which requires ECU to be able to respond quickly after receiving a message and execute the corresponding instructions. Under this premise, the encryption technique should also not affect the transmission rate of the CAN-FD bus. In order to ensure the normal use of CAN-FD bus, we choose a symmetric encryption algorithm with high efficiency of encryption and decryption, small amount of encryption calculation, fast speed, and suitable for encrypting a large number of data. Symmetric encryption algorithm uses the same Key in the encryption and decryption process, which is usually called the "Session Key". This encryption technology is widely used in the field of information security.

Among many encryption algorithms, the AES encryption algorithm has the characteristics of fast operation speed and high security. However, the vehicle electronic control system is implemented in the ECU, so the AES-128 technology based on complex algorithm is difficult to be directly applied to the hardware system of the vehicle electronic control unit Therefore, the AES-128 encryption algorithm used in the proposed security protocol calculates the 128-bit random data value and then carries out XOR operation with the data in the dynamic data compression algorithm to ensure the confidentiality and real-time performance of the message.

C. Certification technology

Identity authentication technology is an effective method to identify operator identity in the computer network. As the key node of network protection, identity authentication plays a pivotal role. Message authentication is one of the commonly used authentication technologies. Message authentication is to verify the integrity of the message in normal transmission, that is, to distinguish whether the message received by the receiver is true or has not been modified. It has two meanings: one is to ensure that the source of the message sender is not an impostor sending an abnormal message; The second is to verify whether the data is modified, replayed or delayed during the transfer or storage process. First, the message is given an authentication code through a specific hash algorithm, which is then sent to the receiver. The receiving end obtains an authentication code through an operation using the conventional algorithm. When the receiving packet receives the authentication codes of the sender, it compares the authentication codes of the sender and the receiver. If the two are equal, the information is complete and accurate; otherwise, the information is modified.

There are three methods for message authentication: One is to encrypt the message and use the encrypted ciphertext as the discriminator. The second is to use a message authentication code (MAC), which takes the message and key as input, and generates a fixed length value as the discriminator. The third is a hash function, also called a hash function or hash function, which uses a hash value as its discriminator. In recent years, message authentication methods are more inclined to use hash functions to design message authentication codes. One is the hash function execution speed is fast, the other is the encryption hash function open source wide.

HMAC is a key-dependent Hash-based Message Authentication Code. HMAC takes a message and a key as input, and generates a message digest of a specific length as output. Among them, the hash algorithm can use the existing hash algorithm on the premise that it only needs to deal with the key. HMAC algorithm can directly apply the existing hash function, and can maintain the performance of the original hash function, will not be too low because of embedding, the use and processing of the key are also relatively simple. Now, the HMAC algorithm has withstood the test of various forms of attacks and become the security standard of the Internet. So this paper uses the HMAC algorithm, which is more like an encryption algorithm, it introduces a key, its security is not completely dependent on the hash function used, so the security and real-time will be higher.

D. Experiment and analysis

The safety protocol model of CAN-FD bus was constructed in vehicle bus simulation software CANoe, and the data transmitted by CAN-FD bus during normal running of several vehicles were collected and replayed in the simulation network, so as to reproduce the CAN bus scene in the process of vehicle implementation and realize the proposed safety protocol. In order to test whether the proposed CAN-FD bus security protocol CAN guarantee the security of CAN-FD bus, three attack modes of CAN-FD bus were simulated in CANoe software for

Authorized licensed use limited to: Notre Dame University - Louaize. Downloaded on September 21,2026 at 19:39:41 UTC from IEEE Xplore. Restrictions apply.

intrusion testing, and the security of the proposed CAN-FD bus security protocol was evaluated.

1) Snooping Data. When there is no security protocol inside the CAN bus, the attacker can use OBD and other tools to access the "plaintext" data which can be easily transmitted in real time inside the CAN bus When the proposed security protocol is implemented in the CAN bus, the original transmitted data changes from regular "plaintext" to irregular "ciphertext". The attacker can't reverse out useful data from irregular "ciphertext" data. According to the proposed security protocol, the "ciphertext" can be restored to the original message.

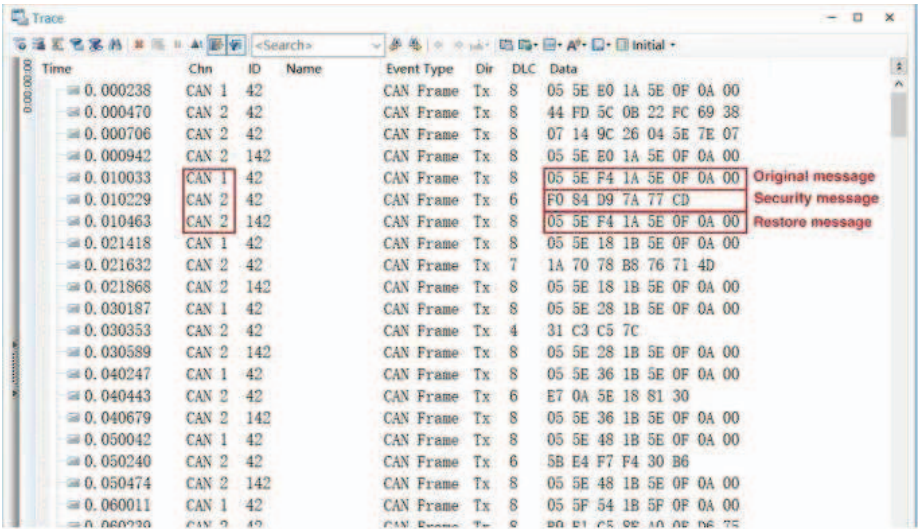
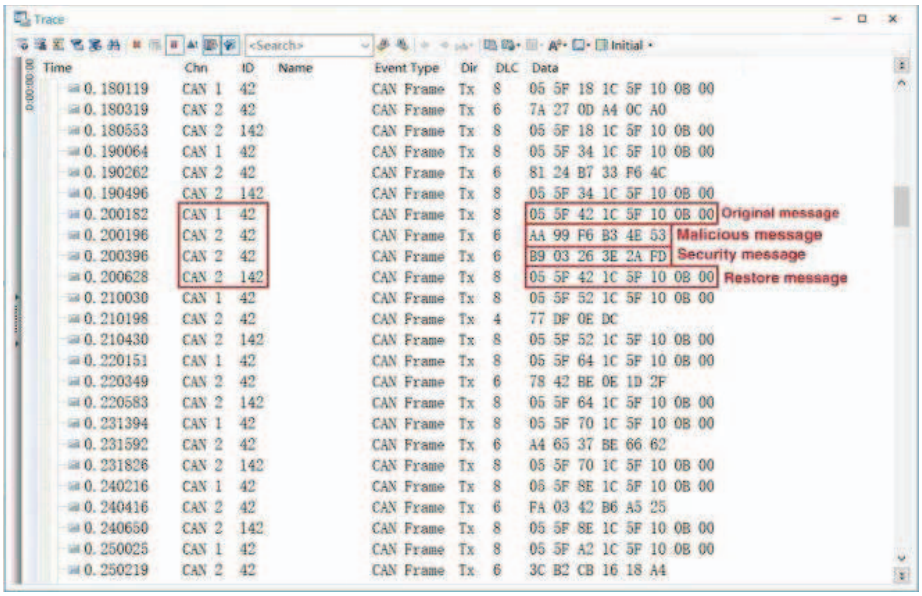


Figure 3. Result chart of message simulation

2) Injecting Malicious Data. When a malicious message is injected into the CAN-FD bus, the malicious message injected by the CAN-FD bus broadcast mechanism is executed on the corresponding ECUs. However, when ECU implements the CAN-FD bus security protocol proposed by ECU, although ECU with the corresponding ID will also receive malicious messages, the HMAC authentication corresponding to malicious messages cannot pass due to HMAC authentication in the proposed security protocol, so ECU will discard this malicious message. Thus, malicious messages cannot affect the normal execution of ECUs. As shown in Figure 4, when the "malicious message" is injected into the CAN-FD bus, the ECU will still receive the malicious message. However, the malicious message will not affect the normal verification, decryption, and decompression of the "secure message" at the receiving end because it cannot pass HMAC data authentication.



Authorized licensed use limited to: Notre Dame University - Louaize. Downloaded on September 21,2026 at 19:39:41 UTC from IEEE Xplore. Restrictions apply.

Figure 4. Result chart of simulation when injected malicious message

3) Flooding Data. When attack messages are repeatedly injected into the CAN-FD bus that does not implement security protocols, the attacked ECUs will receive two types of messages: messages sent by normal ECU nodes and malicious attack messages. After receiving the two types of messages, the ECUs will perform operations based on the two types of messages. However, when the ECU implements the CAN-FD bus security protocol proposed, although the ECU will receive normal messages and malicious messages, the malicious messages cannot pass the HMAC authentication of the security protocol proposed, so the ECU will not execute malicious messages but only execute normal messages. Figure 5 shows the rotational speed of the vehicle engine during the flood data attack. Where Dr_N represents the engine speed when the vehicle is running normally; N represents the speed of the attacked engine without safety measures; UA_N indicates the engine speed under attack when security measures exist. For example, when the vehicle is running normally, the engine speed decreases from about 4300 rpm to about 1900 rpm during the first 6 seconds of acceleration, and then decreases all the time, as shown in Dr_N. If the malicious message is injected again at the sixth second, the engine speed will be affected by the malicious message. The attacked ECUs will receive both normal and malicious messages, which will cause the engine speed it receives to fall between normal and malicious values, as shown by the "N" in the figure. When the corresponding ECU implements the security protocol mentioned above, the malicious message is also injected at the 6th second. Although the ECU will still receive the malicious message, the malicious message cannot pass HMAC verification, so the engine speed of the vehicle is still the same as that in normal running, as shown in "UA_N" in the figure.

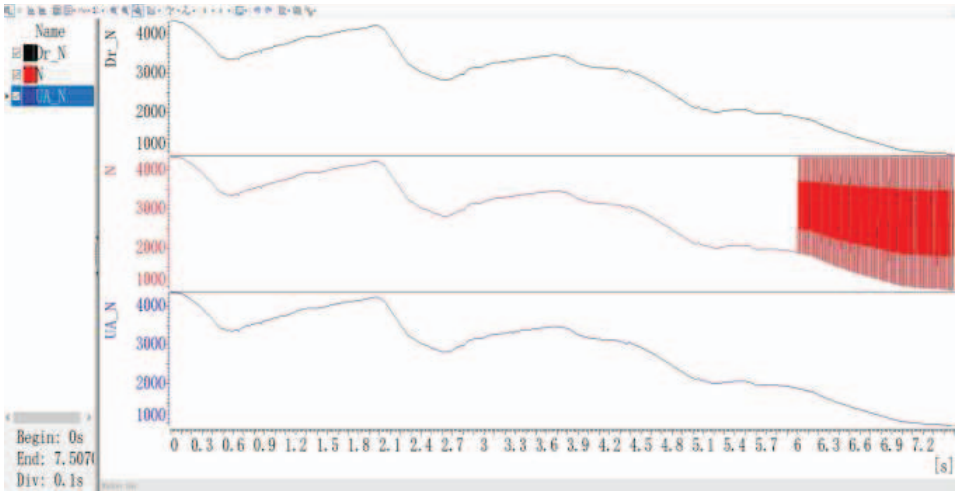


Figure 5. Comparison chart of engine speed when flooding dat

V. CONCLUSIONS

This paper puts forward the importance of protecting vehicle information security, and puts forward a practical security scheme. After understanding the network security problems faced by automobile network, the framework of CAN-FD is analyzed in detail. Secondly, we summarize the possible vulnerabilities of CAN-FD and four possible attack types. Finally, according to the possible attack types, the detection data anomaly and encryption data authentication technologies are proposed, and a feasible scheme is proposed to protect the vehicle CAN-FD bus.

ACKNOWLEDGMENT

This research was supported by National Natural Science Foundation of China (grant number 62161049 and 62201492). The Science & Technology Development Project of Jilin Province (Grant No.20220101141JC). National 2023 Innovation Training Program for University Students "Intelligent Networked Vehicle CAN/CAN-FD Bus Network Security".

REFERENCES

[1] Y.Xun, Y.Sun, J.Liu.(2019) An experimental study towards driver identification for intelligent and connected vehicles. International Conference on Communications.

[2] J.Xu, J.Wang, C.Liu, L.Zhou, F.Long. (2022) Cyber security and anomaly detection method for in-vehicle CAN. Journal Of Shandong University.

[3] J.Fang. (2022) Intrusion detection and fault tolerance control of CAN-FD bus network. Research, 12-17.

[4] T.Song, H.Li, H.Li, J.Wen, Q.Xiong, J.Chen. (2022) Overview of Research on Security Encryption Authentication Technology of IoV in Big Data Era. Engineering Technology II Series, 340-453.

[5] B.Groza, S.Murvay, A.Herrewege, I.Verbauwhede. (2012) LiBrA-CAN: A Lightweight Broadcast Authentication Protocol for Controller Area Networks. Cryptology and Network Security, 185-200.

[6] B.Groza, S.Murvay. (2013) Efficient Protocols for Secure Broadcast in Controller Area Networks. IEEE Transactions on Industrial Informatics.

[7] Ding Xiangzhe X,Ding. (2021) Design of CAN BUS security protocol based on dynamic data compression algorithm.Research, 45-46.

Authorized licensed use limited to: Notre Dame University - Louaize. Downloaded on September 21,2026 at 19:39:41 UTC from IEEE Xplore. Restrictions apply.